



**Empowering Hidden Creators -
Predicting YouTube Video Popularity
from Metadata**

Jayden Doan

Problem



Many creators fear posting videos that won't be seen (wasted effort)



Popular creators struggle to predict what will trend



Built a ML model to predict video popularity using metadata (before upload)

Stakeholders



Niche/New Creators: Want confidence to publish and test bold ideas



Popular Creators: Need help validating and prioritising content

Value



Small view increases = real money



 +10,000 views = \$50-\$300 per video



 Weekly uploads = \$200-\$1,200/month extra



 At scale: +1M views = \$5,000-\$30,000 gain

Data - Youtube Trending Videos Dataset

- video_id
- video_published_at
- video_trending_date
- video_trending_country
- video_title
- video_description
- video_default_thumbnail
- video_category_id
- video_tags
- video_duration
- video_dimension
- video_definition
- video_licensed_content.
- video_view_count
- video_like_count
- video_comment_count
- channel_id
- channel_title
- channel_description
- channel_custom_url
- channel_published_at
- channel_country
- channel_view_count
- channel_subscriber_count
- channel_have_hidden_subscribers
- channel_video_count
- channel_localized_title
- channel_localized_description
- <https://www.kaggle.com/datasets/canerkonuk/youtube-trending-videos-global>
- 28 columns, 4,165,967 rows
- Updated daily

Data Cleaning

Drop duplicates of 'video_id'

Drop rows where 'video_view_count' was missing

Drop rows where 'channel_subscriber_count' was missing

Rows: 4,165,967 -> 148,985

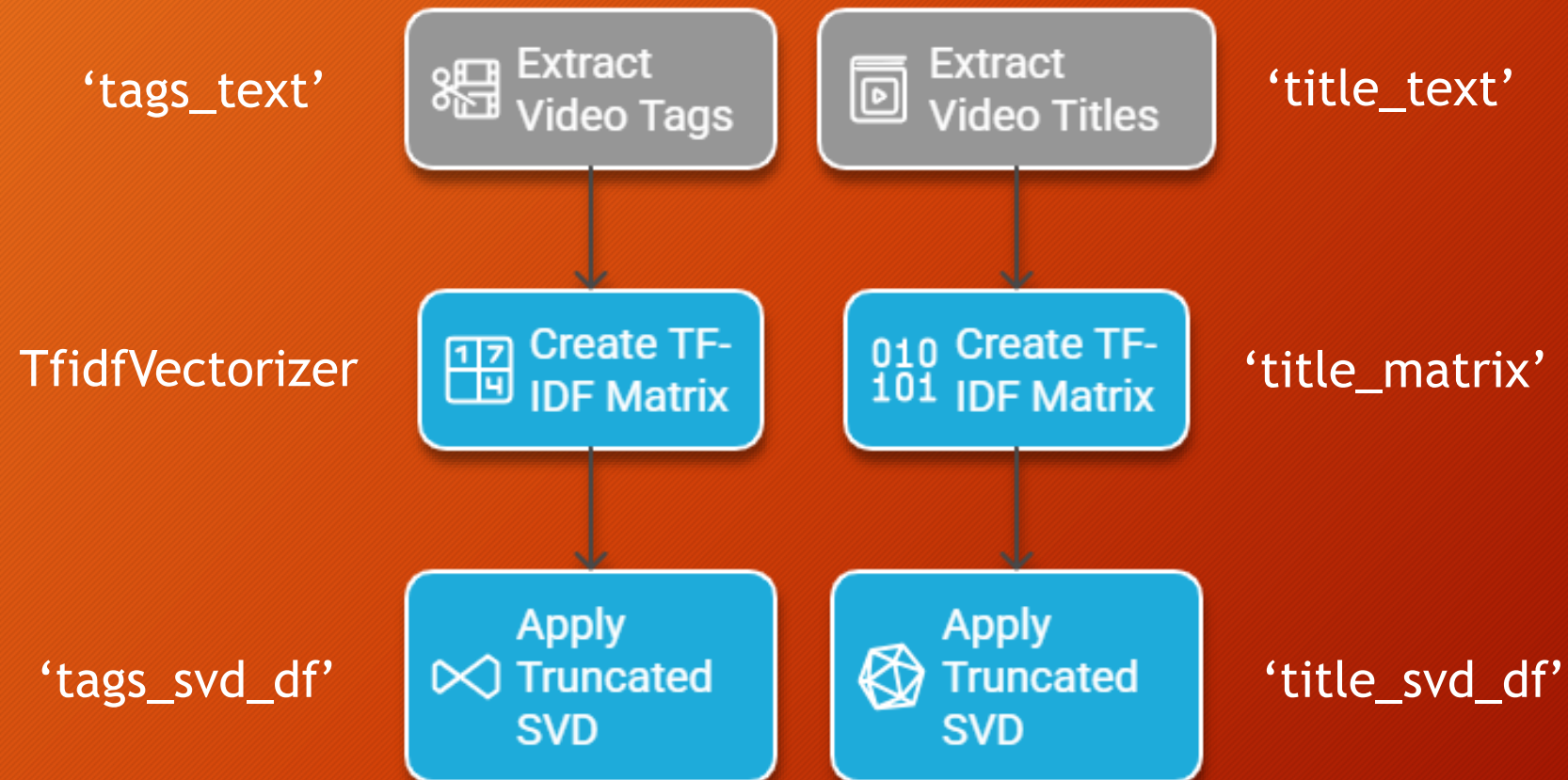
Feature engineering

- 'video_duration_sec'

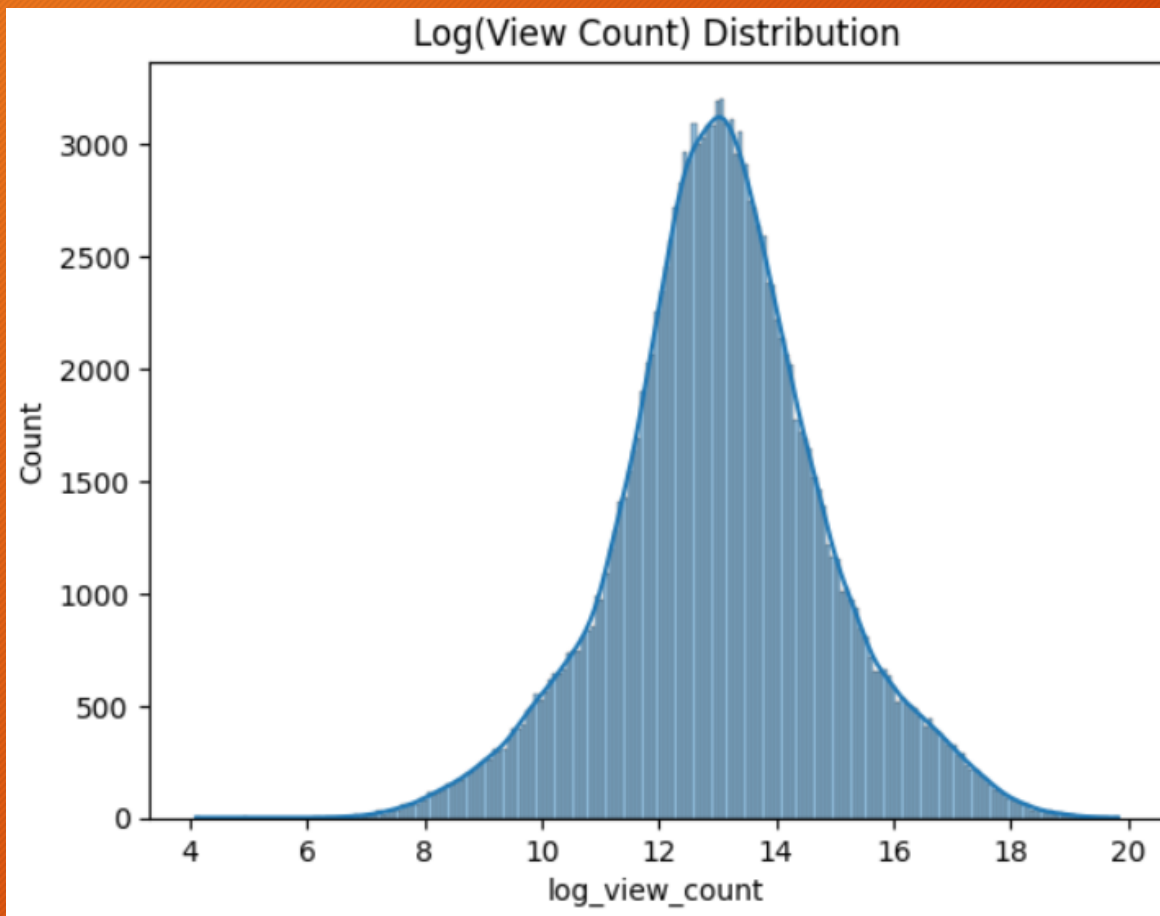
	video_duration	video_duration_sec
10173	PT2M31S	151.0
10183	PT11M54S	714.0
9982	PT40M11S	2411.0
10021	PT9M48S	588.0
10026	PT24S	24.0

- Convert 'video_definition' to binary column 'is_hd'
- 'title_length' using 'video_title'
- 'tag_count' using 'video_tags'
- 'log_view_count' using 'video_view_count'

Feature engineering

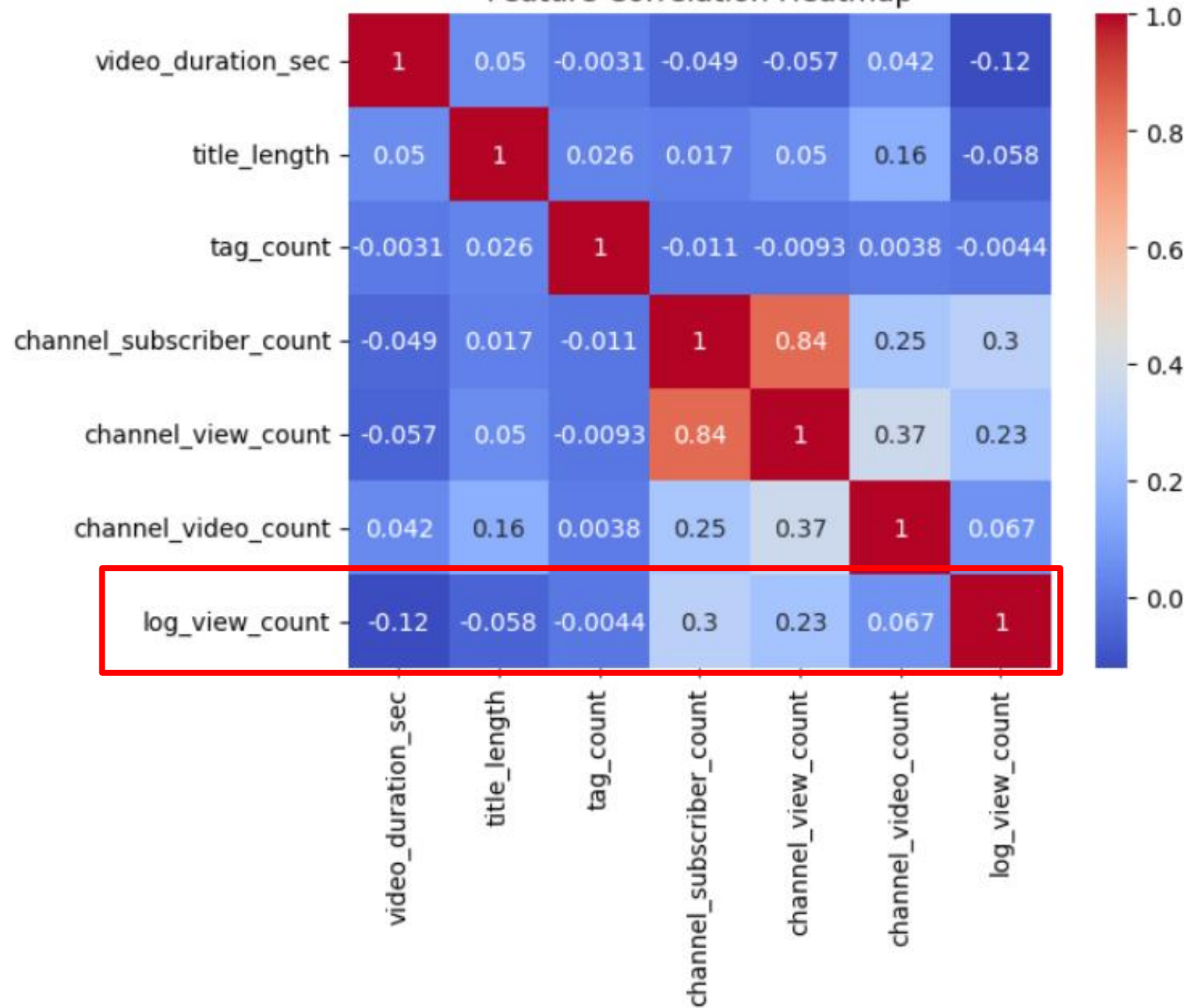


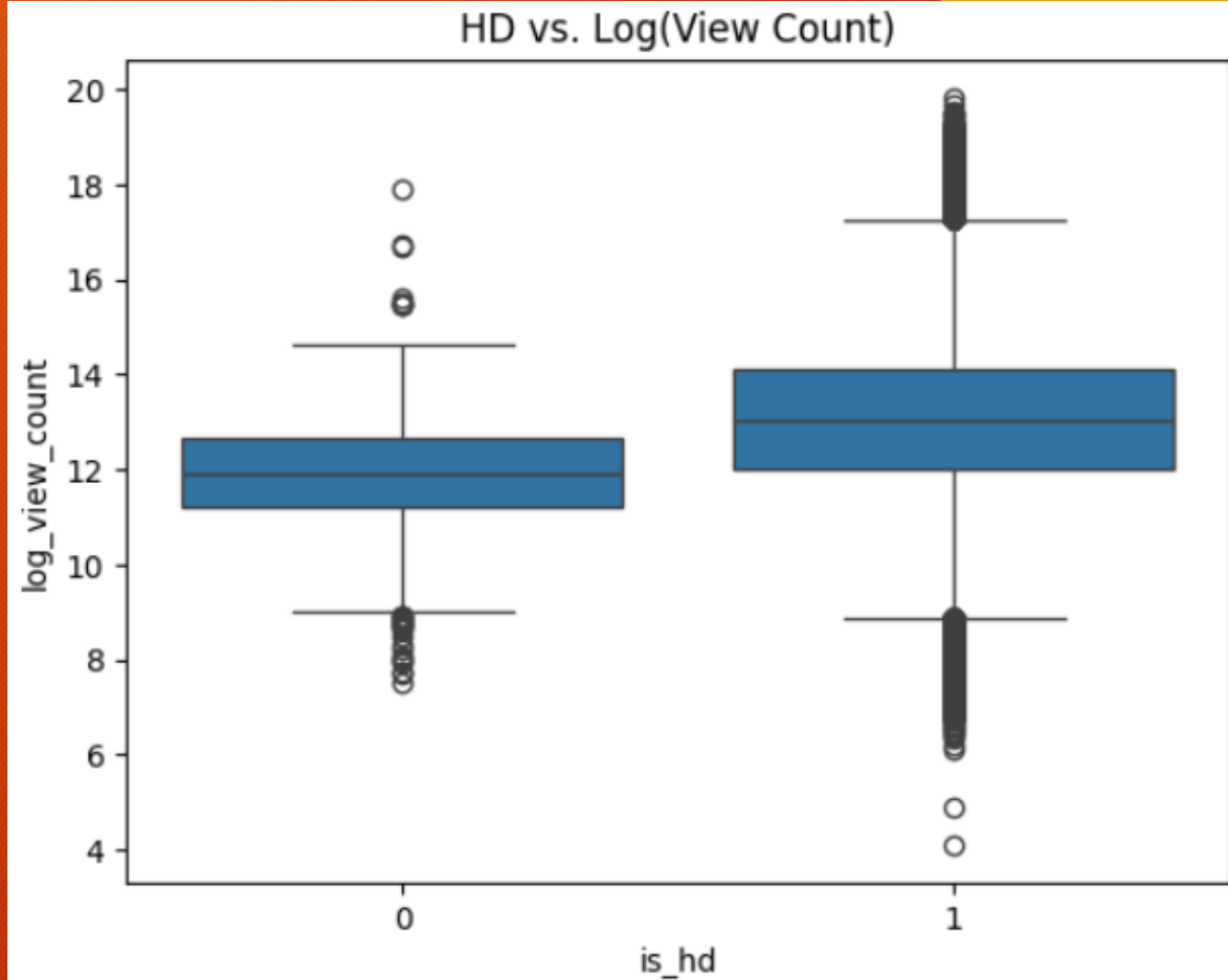
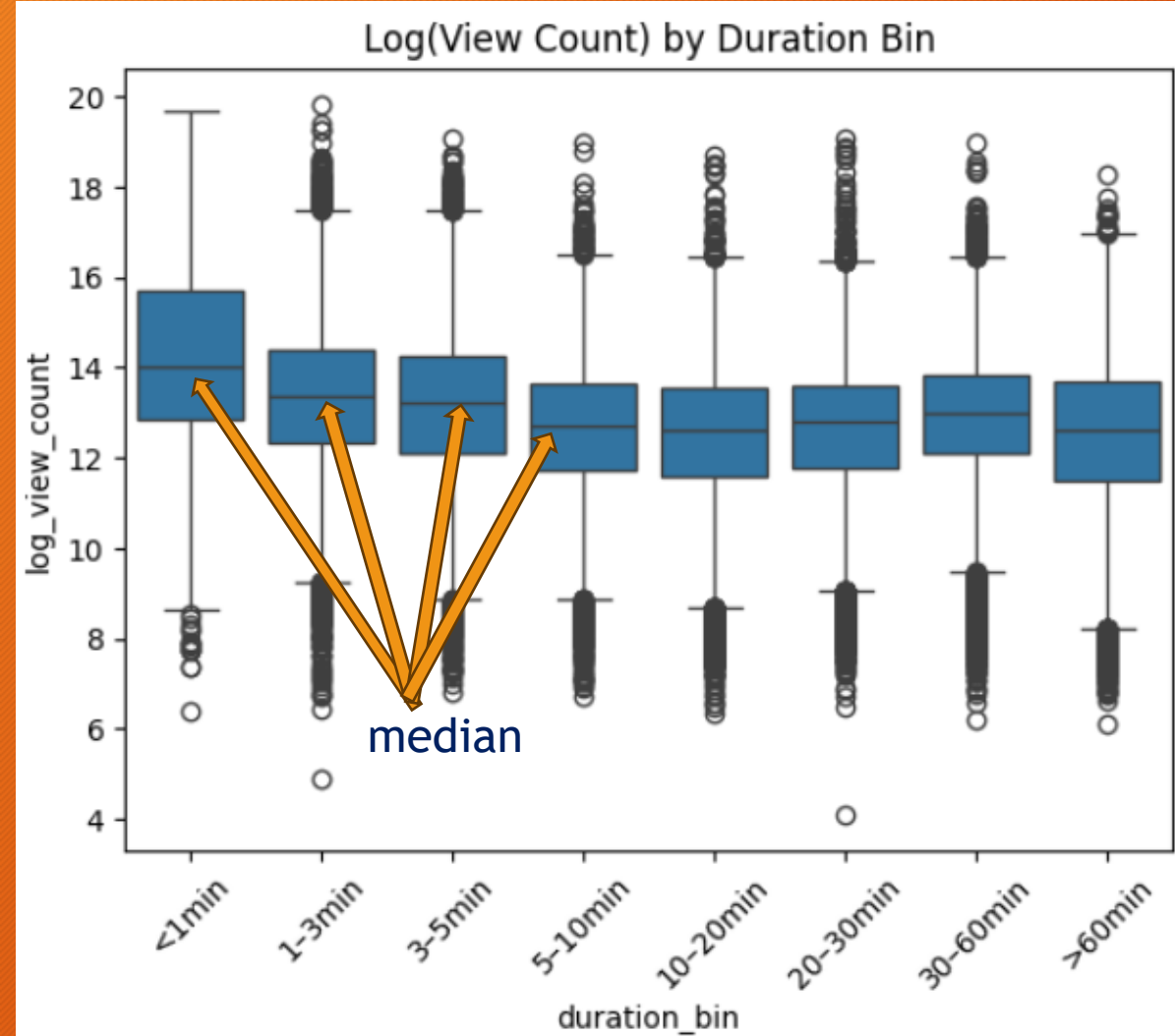
Exploratory Data Analysis



- Min: $\text{Exp}(4) - 1 \approx 54$ views
- $\text{Exp}(20) - 1 \approx 485,000,000$ views
- $\text{Exp}(13) - 1 \approx 442,000$ views

Feature Correlation Heatmap





Modelling

```
selected_features = [  
    'video_duration_sec',  
    'title_length',  
    'is_hd',  
    'channel_subscriber_count',  
    'channel_view_count',  
    'channel_video_count',  
] + list(tags_svd_df.columns) + list(title_svd_df.columns)
```

```
X = df_tfidf[selected_features]  
y = df_tfidf['log_view_count']
```


Modelling

Gradient Boosting

Builds models sequentially to correct errors.

Linear Regression

Models linear relationships between variables.

Random Forest

Combines multiple decision trees for accuracy and overfitting control.



XGBoost

Optimized gradient boosting for speed and performance.

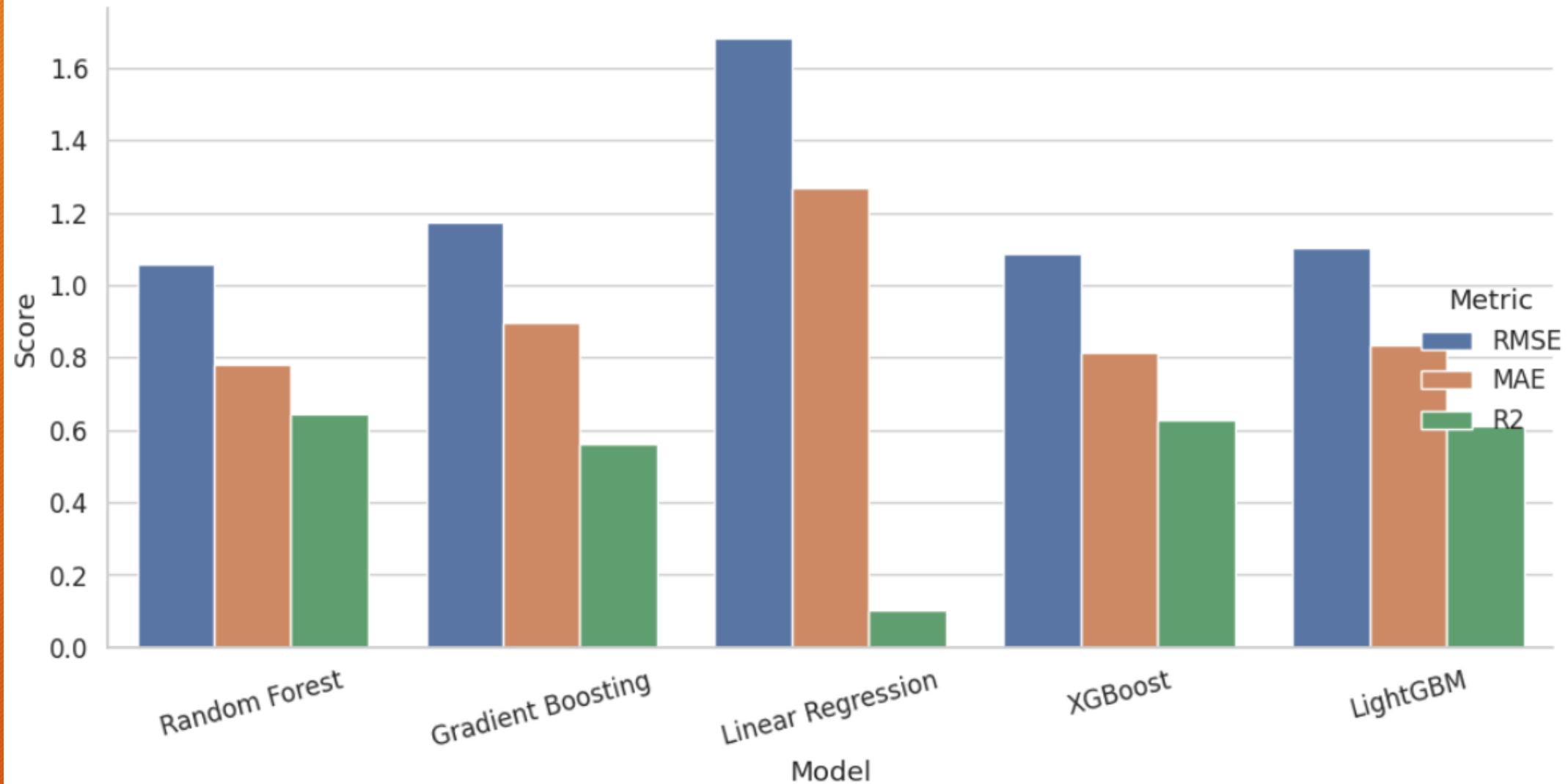
LightGBM

Efficient and scalable gradient boosting for large datasets.

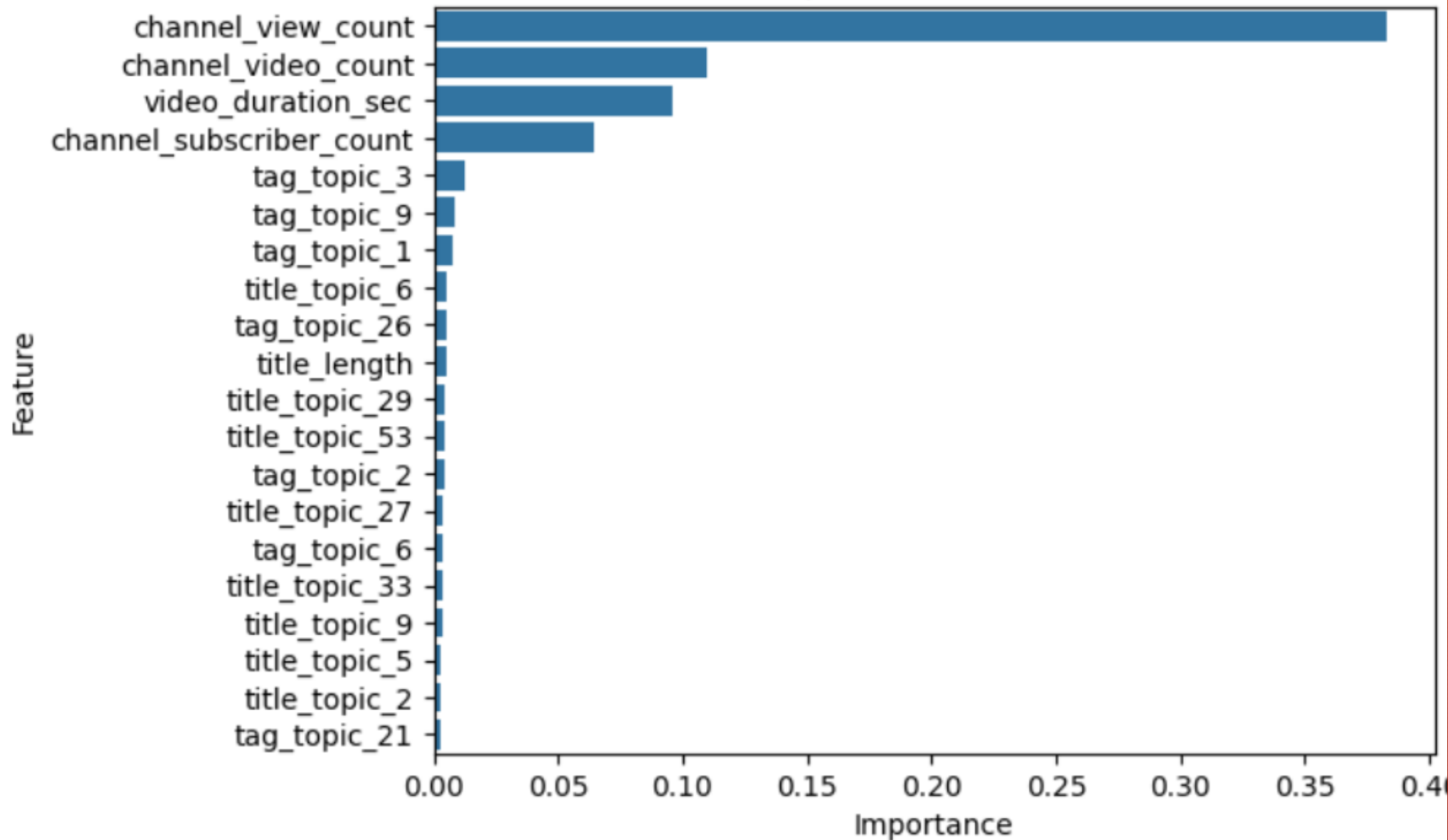
Model Comparison

Model	Random Forest	Linear Regression	Gradient Boosting	XGBoost	LightGBM
RMSE	1.049	1.690	1.170	1.079	1.096
MAE	0.776	1.281	0.896	0.813	0.829
R ² Score	0.656	0.105	0.571	0.635	0.624
MAE (% error)	±118.5%	±260.0%	±145.0%	±125.5%	±129.1%

Model Performance Comparison



Feature Importance (Random Forest)



Key Takeaways for Stakeholders



PREDICTED VIDEO
POPULARITY USING
ONLY METADATA.



SMALL
IMPROVEMENTS CAN
DRIVE BIG VALUE



CREATIVE IDEAS CAN
BE DE-RISKED



MODEL IS CREATOR-
FRIENDLY AND
SCALABLE

Limitations

Time sensitivity for trends

Bias toward established channels

Limited generalisation to original content

Metadata-only input

Thank you!